- **Internship Name : Data Analysis**
- **Task Name: Task 8 : Importing And Exporting Datasets using Pandas**
- **Your Name : Shubhangi Naresh Khonde**
- **Internship Organization :**
- **Date:**

1. Introduction

In data analysis, data is usually stored in files such as CSV and Excel sheets. Pandas provides simple functions to read data from these files into a DataFrame and to save the results back into files. In this task, I practiced the four main Pandas input/output functions: read_csv(), read_excel(), to_csv() and to_excel(), using a real employee dataset.

2. Objectives

- To import a CSV file into a Pandas DataFrame using read_csv()
- To export a DataFrame to an Excel file using to_excel()
- To import an Excel file into a DataFrame using read_excel()
- To export filtered data to a CSV file using to_csv()
- To verify that the exported files are created and readable

3. Tools and Technologies Used

- **Language:** Python 3.13
- **Libraries:** Pandas, openpyxl (for reading/writing Excel files), os
- **Environment:** Jupyter Notebook in Visual Studio Code
- **Dataset:** Excel_Internship_Task.csv (employee records)

4. Dataset Description

The dataset contains 2500 employee records (rows) and 14 columns:

Column	Description
Employee_ID	Unique ID of the employee
Employee_Name	Name of the employee
Gender	Male / Female
Age	Age of the employee
Department	Department such as Sales, IT, HR, Finance, Marketing
Job_Title	Designation of the employee
City	City of work
State	State of work
Joining_Date	Date of joining
Salary	Salary of the employee
Performance_Score	Performance score
Bonus	Bonus amount
Employment_Status	Active / Inactive / On Leave
Grade	Grade of the employee

5. Methodology and Implementation

Step 1: Importing Libraries

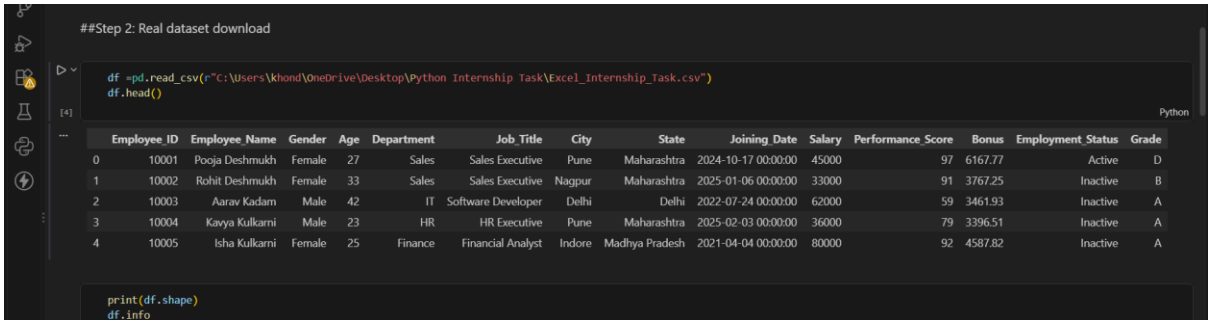
Pandas was imported as pd. The openpyxl library was installed because Pandas uses it internally to read and write .xlsx files.

```
!pip install openpyxl
import pandas as pd
```

Step 2: Importing a CSV File using read_csv()

The employee dataset was loaded from a CSV file into a DataFrame, and head() was used to preview the first 5 rows.

```
df = pd.read_csv(r"C:\Users\khond\OneDrive\Desktop\Python Internship Task\Excel_Internship_Task.csv")
df.head()
```



```
##Step 2: Real dataset download

df =pd.read_csv(r"C:\Users\khond\OneDrive\Desktop\Python Internship Task\Excel_Internship_Task.csv")
df.head()
```

Python

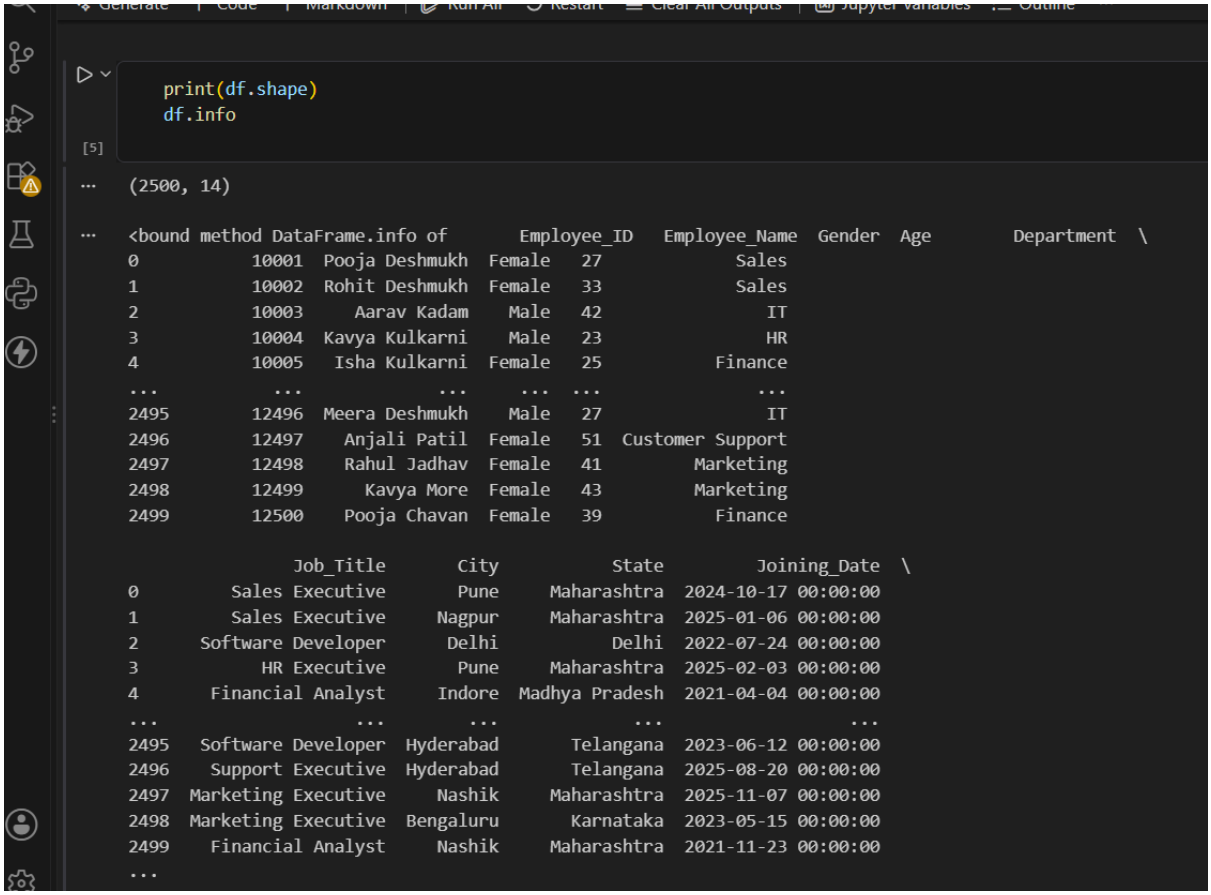
	Employee_ID	Employee_Name	Gender	Age	Department	Job_Title	City	State	Joining_Date	Salary	Performance_Score	Bonus	Employment_Status	Grade
0	10001	Pooja Deshmukh	Female	27	Sales	Sales Executive	Pune	Maharashtra	2024-10-17 00:00:00	45000	97	6167.77	Active	D
1	10002	Rohit Deshmukh	Female	33	Sales	Sales Executive	Nagpur	Maharashtra	2025-01-06 00:00:00	33000	91	3767.25	Inactive	B
2	10003	Aarav Kadam	Male	42	IT	Software Developer	Delhi	Delhi	2022-07-24 00:00:00	62000	59	3461.93	Inactive	A
3	10004	Kavya Kulkarni	Male	23	HR	HR Executive	Pune	Maharashtra	2025-02-03 00:00:00	36000	79	3396.51	Inactive	A
4	10005	Isha Kulkarni	Female	25	Finance	Financial Analyst	Indore	Madhya Pradesh	2021-04-04 00:00:00	80000	92	4587.82	Inactive	A

```
print(df.shape)
df.info
```

Step 3: Understanding the Dataset

The shape of the DataFrame and its structure were checked. The shape was (2500, 14), meaning 2500 rows and 14 columns. The info() function shows the column names, non-null counts and data types.

```
Print(df.shape)
df.info()
```



```
print(df.shape)
df.info
```

[5]

```
(2500, 14)

<bound method DataFrame.info of       Employee_ID   Employee_Name  Gender  Age       Department  \
0           10001  Pooja Deshmukh  Female   27            Sales
1           10002  Rohit Deshmukh  Female   33            Sales
2           10003     Aarav Kadam    Male   42               IT
3           10004  Kavya Kulkarni    Male   23               HR
4           10005   Isha Kulkarni  Female   25          Finance
...           ...             ...     ...  ...              ...
2495        12496  Meera Deshmukh    Male   27               IT
2496        12497    Anjali Patil  Female   51  Customer Support
2497        12498    Rahul Jadhav  Female   41        Marketing
2498        12499      Kavya More  Female   43        Marketing
2499        12500    Pooja Chavan  Female   39          Finance

                Job_Title       City          State        Joining_Date  \
0         Sales Executive       Pune    Maharashtra 2024-10-17 00:00:00
1         Sales Executive     Nagpur    Maharashtra 2025-01-06 00:00:00
2      Software Developer      Delhi          Delhi 2022-07-24 00:00:00
3            HR Executive       Pune    Maharashtra 2025-02-03 00:00:00
4       Financial Analyst     Indore  Madhya Pradesh 2021-04-04 00:00:00
...                   ...        ...             ...                 ...
2495   Software Developer  Hyderabad       Telangana 2023-06-12 00:00:00
2496    Support Executive  Hyderabad       Telangana 2025-08-20 00:00:00
2497  Marketing Executive     Nashik    Maharashtra 2025-11-07 00:00:00
2498  Marketing Executive  Bengaluru       Karnataka 2023-05-15 00:00:00
2499    Financial Analyst     Nashik    Maharashtra 2021-11-23 00:00:00
...
```

Step 4: Exporting to Excel using to_excel()

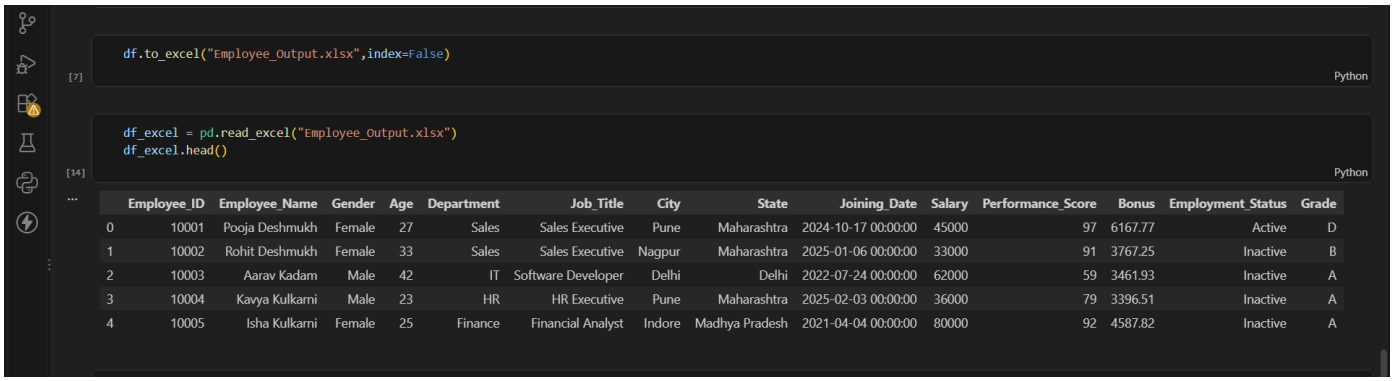
The complete DataFrame was saved as an Excel file. The parameter index=False prevents the row index from being saved as an extra column.

```
df.to_excel("Employee_Output.xlsx", index=False)
```

5: Importing an Excel File using read_excel()

The exported Excel file was read back into a new DataFrame to confirm that the data was saved correctly.

```
Df_excel = pd.read_excel(“Employee_Output.xlsx”)
df_excel.head()
```



```
df.to_excel("Employee_Output.xlsx",index=False)
```

Python

```
df_excel = pd.read_excel("Employee_Output.xlsx")
df_excel.head()
```

Python

	Employee_ID	Employee_Name	Gender	Age	Department	Job_Title	City	State	Joining_Date	Salary	Performance_Score	Bonus	Employment_Status	Grade
0	10001	Pooja Deshmukh	Female	27	Sales	Sales Executive	Pune	Maharashtra	2024-10-17 00:00:00	45000	97	6167.77	Active	D
1	10002	Rohit Deshmukh	Female	33	Sales	Sales Executive	Nagpur	Maharashtra	2025-01-06 00:00:00	33000	91	3767.25	Inactive	B
2	10003	Aarav Kadam	Male	42	IT	Software Developer	Delhi	Delhi	2022-07-24 00:00:00	62000	59	3461.93	Inactive	A
3	10004	Kavya Kulkarni	Male	23	HR	HR Executive	Pune	Maharashtra	2025-02-03 00:00:00	36000	79	3396.51	Inactive	A
4	10005	Isha Kulkarni	Female	25	Finance	Financial Analyst	Indore	Madhya Pradesh	2021-04-04 00:00:00	80000	92	4587.82	Inactive	A

Step 6: Exporting Filtered Data to CSV using to_csv()

Instead of saving all data, only the employees whose Employment_Status is Active were filtered and exported to a new CSV file. The file was then read back to verify it.

```
active_emp = df[df["Employment_Status"] == "Active"]
active_emp.to_csv("Active_Employees.csv", index=False)
pd.read_csv("Active_Employees.csv").head()
```



```
active_emp = df[df["Employment_Status"] == "Active"]
active_emp.to_csv("Active_Employees.csv", index=False)
```

Python

```
pd.read_csv("Active_Employees.csv").head()
```

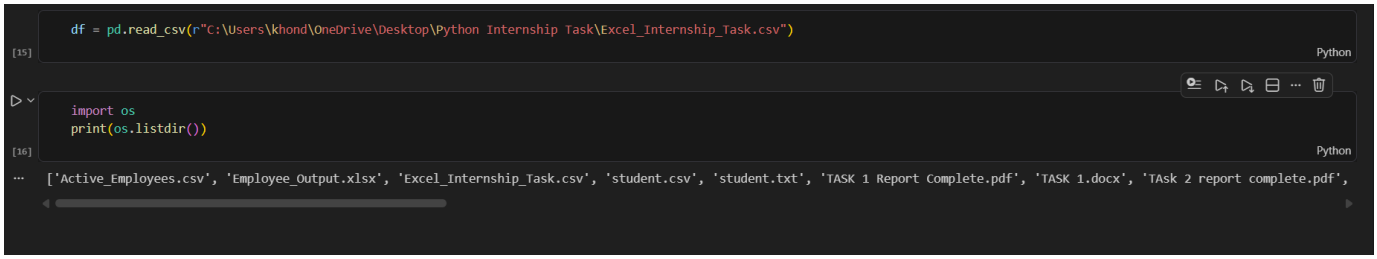
Python

	Employee_ID	Employee_Name	Gender	Age	Department	Job_Title	City	State	Joining_Date	Salary	Performance_Score	Bonus	Employment_Status	Grade
0	10001	Pooja Deshmukh	Female	27	Sales	Sales Executive	Pune	Maharashtra	2024-10-17 00:00:00	45000	97	6167.77	Active	D
1	10010	Kavya Deshmukh	Female	53	HR	HR Executive	Ahmedabad	Gujarat	2023-10-08 00:00:00	55000	55	6638.16	Active	B
2	10013	Neha More	Female	40	IT	Software Developer	Aurangabad	Maharashtra	2025-09-20 00:00:00	93000	90	8455.52	Active	B
3	10016	Meera Kadam	Male	54	Operations	Operations Executive	Mumbai	Maharashtra	2025-11-22 00:00:00	60000	88	4846.59	Active	B
4	10017	Karan More	Male	28	HR	HR Executive	Aurangabad	Maharashtra	2022-05-23 00:00:00	53000	64	1997.20	Active	B

Step 7: Verifying the Created Files

The os module was used to list the files in the working folder. The output showed that Employee_Output.xlsx and Active_Employees.csv were created successfully.

```
import os
print(os.listdir())
```



6. Summary of Functions Used

Function	Purpose	File Used
read_csv()	Import data from a CSV file	Excel_Internship_Task.csv
to_excel()	Export a DataFrame to an Excel file	Employee_Output.xlsx
read_excel()	Import data from an Excel file	Employee_Output.xlsx
to_csv()	Export a DataFrame to a CSV file	Active_Employees.csv

7. Observations

- The dataset has 2500 rows and 14 columns, so it is a good example of a real-world dataset.
- Data exported with to_excel() and to_csv() was read back with the same values, so no data was lost.
- Filtering a DataFrame before exporting allows only the required records to be saved.
- Using index=False keeps the exported files clean, without an extra index column.

8. Learning Outcomes

- I learned how to import CSV and Excel files into Pandas DataFrames.
- I learned how to export DataFrames to Excel and CSV files.
- I understood the importance of using the correct function for each file type.
- I learned how to give file paths in Python and how to check files using the os module.
- I learned how to read common errors and fix them.

9. Conclusion

This task gave me practical experience in working with real datasets in Pandas. I successfully imported a CSV file, exported data to Excel, imported the Excel file, and exported filtered data to CSV. These input/output skills are essential for any data analysis workflow, because data is usually received and shared in these file formats.